



Subtitle Generation System

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Technologies Used

- Programming Lang – Python, Java
- Cloud – MS Azure
- Cloud Services/Features – Virtual Machine, Translator, PostgreSQL DB
- Browser – Chrome
- API's – Whisper
- IDE – VS Code, Eclipse
- Communication – JSON
- Servers – FastAPI, Apache Tomcat 9.0
- OS: Linux (Ubuntu 24.04 LTS), Windows 11 Pro

Model & Cost Mapping

Component	Cloud Service	Service Model	Purpose	Example Configuration	Approx Cost
Speech Recognition (Whisper ASR)	Azure Virtual Machine	IaaS	Hosts Python Whisper server for speech-to-text processing	Standard D4s v3 (4 vCPU, 16GB RAM)	~\$0.19/hour (~\$140/month if running 24/7)
Subtitle Database	Azure Database for PostgreSQL	PaaS	Stores metadata, subtitles, and translations	Flexible Server – 2 vCore, 8GB RAM	~\$50–\$70/month
Text Translation	Azure AI Translator	SaaS	Translates subtitles from source language to English	Translator Text API	~\$10 per 1M characters
Application Backend	Java Servlet (Tomcat)	-	Handles file upload, calls Whisper API, manages translation and DB	Hosted on VM or App Service	Included in VM cost
Frontend	JSP + JavaScript	Web App	Displays subtitles and allows SRT download	Hosted with backend	Included

Resource	Usage Example	Estimated Cost
Azure VM	6 hours/day usage	~\$34/month (\$140/month)
PostgreSQL	Basic Flexible Server	~\$50/month (\$75/month)
Azure Translator	200K characters/month	~\$2 (\$10 per 1M chars)
Total Estimated Monthly Cost		\$86/month

Project Concepts

- Automatic Speech Recognition (ASR) using Whisper
- Language Translation using Azure AI Translator
- REST API communication between Java and Python FastAPI using JSON
- Asynchronous processing and background tasks using AJAX
- Azure Virtual Machine and PostgreSQL DB
- Multithreading and Thread Pooling

System Architecture

- Java Web Application (Servlet + JSP UI)
- Python FastAPI server running Whisper ASR
- Azure Translator for subtitle translation
- Azure PostgreSQL for storing metadata and subtitles
- UI displays subtitles and allows '.srt' file download

Deployment Architecture

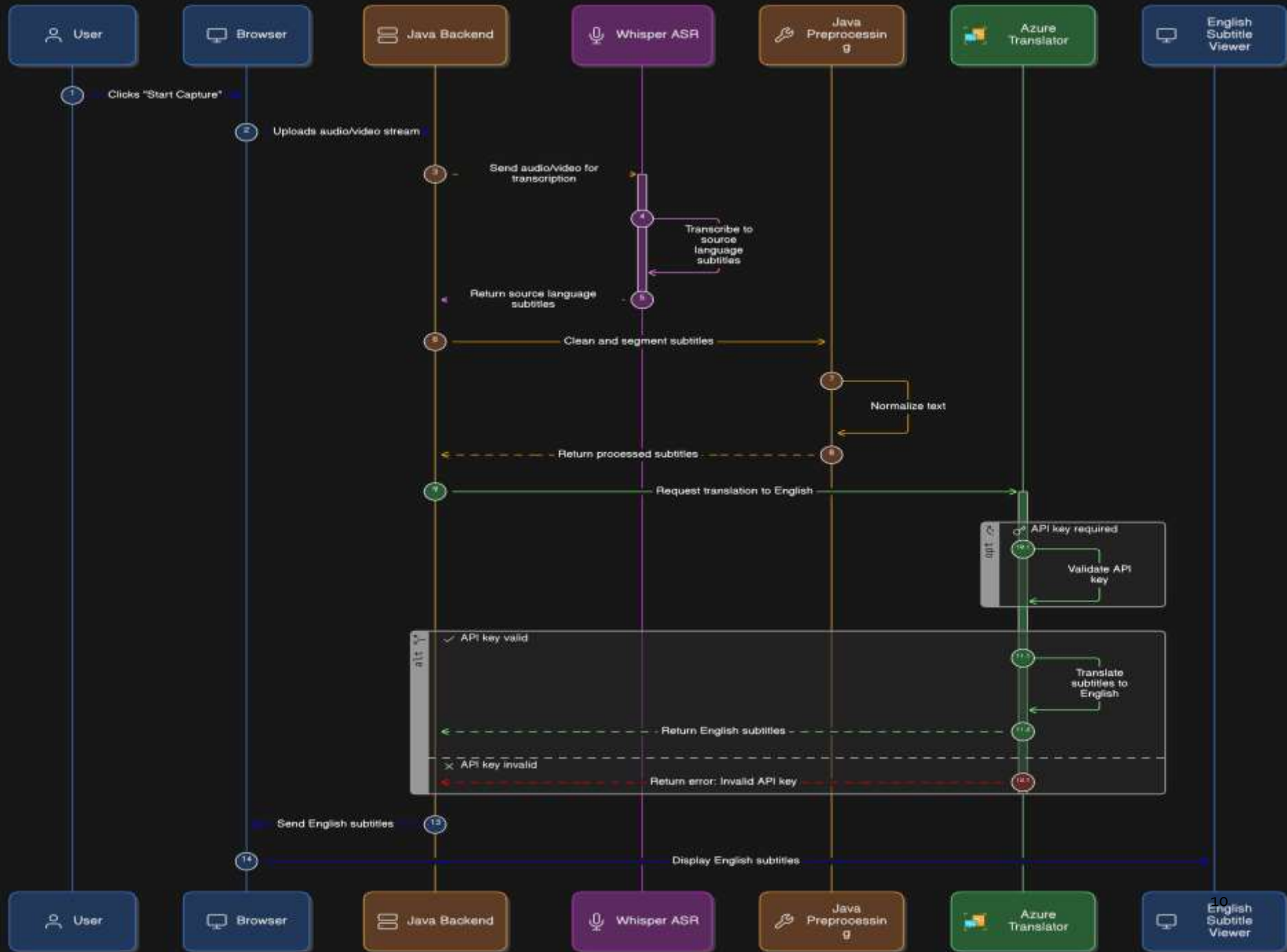
- Whisper ASR server deployed on Azure Virtual Machine
- Java web application deployed on Apache Tomcat
- Database hosted on Azure PostgreSQL
- Communication via REST API over HTTP
- Cloud infrastructure enables remote access and scalability

Cloud Services Used

- Azure Virtual Machine (IaaS)
- Azure AI Translator (SaaS)
- Azure Database for PostgreSQL (PaaS)
- Azure Networking – firewall rules



Project Workflow



#	Start	End	Subtitle
1	00:00:00,000	00:00:04,059	One of the most interesting things I read was still believed all over the world.
2	00:00:04,059	00:00:08,400	The biggest snake was found in the US and he
3	00:00:08,400	00:00:12,400	The name Titana Bua was some 8-9 meters long.
4	00:00:14,220	00:00:18,400	The King of Snakes was released just over a year ago.
5	00:00:18,400	00:00:22,440	Even more than that, the snake found its offspring in Gujarat, 15 meters 15.
6	00:00:22,440	00:00:24,680	The meter is about 50 feet long.
7	00:00:25,000	00:00:29,240	And what was his name? Taitana was named Boga. It has been named Varsuki
8	00:00:29,240	00:00:33,540	Indicus. Now let's reverse this one. Think about it.
9	00:00:33,540	00:00:37,720	If Varsuki's size is 50 feet, then it will also weigh the same as the Saab. Many
10	00:00:37,720	00:00:41,860	100 kg would be a 50-foot snake. So imagine if the annual 50 feet
11	00:00:41,860	00:00:46,000	So how big was the shoulder of the one on whose neck they flowed?
12	00:00:46,000	00:00:49,680	Will be. And his shoulder was that big, so what would be Schiff's size then?
13	00:00:50,000	00:00:54,160	And if Shiva's size was the same, then the vehicle on which Shiva comes is called Nandi.
14	00:00:54,160	00:00:58,340	What would it be like to carry both Shiva and Vasuki to his own devices?
15	00:00:58,340	00:00:58,860	is.

Conclusion

Observations: Azure cloud offers multiple services, but it is not very easy to use, UI is very slow as compared to Azure CLI.

AWS is definitely comparatively better and easy to use than Azure. Cloud provides ease of deployment post development, but integration during development and integration of multiple cloud services while ensuring smooth workflow of data is difficult.